

AOS Project-2

Linux Web-Server

Team: 3

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Project Demo Link:

<https://umsystem.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=81271b8a-a6d9-425e-b130-b2c100231a2d>

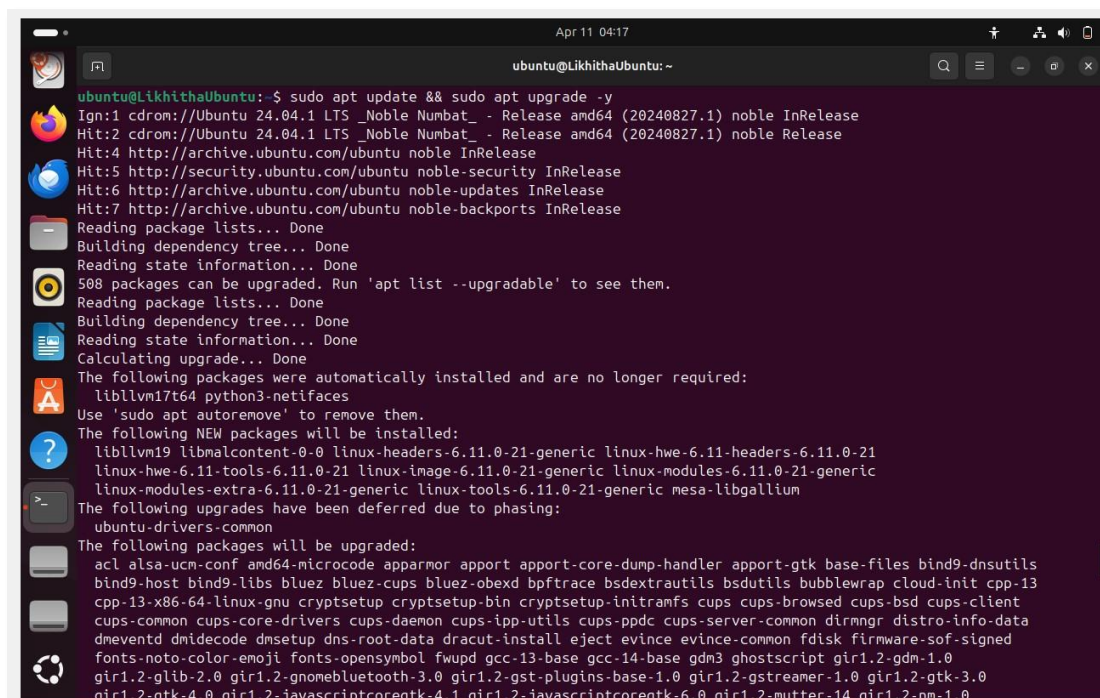
Project: Building a Web Server for NAS Servers Objective:

Create a web-based interface to manage and access your Network Attached Storage (NAS) server. This interface will allow you to perform various tasks remotely, such as:

- **File Management:** (Upload, download, and delete files. Create, rename, and delete folders.)
- **User Management:** (Create, modify, and delete user accounts. Assign permissions to users (read, write, edit, admin))
- **System Monitoring:** (Monitor disk usage, CPU usage, and system logs.) • **Backup and Restore:** (Schedule automatic backups. Restore files and system configurations)

Task:

1.Set up the Linux Server:



```
ubuntu@LikhithaUbuntu: ~  
$ sudo apt update && sudo apt upgrade -y  
Ign:1 cdrom://Ubuntu 24.04.1 LTS _Noble Numbat_ - Release amd64 (20240827.1) noble InRelease  
Hit:2 cdrom://Ubuntu 24.04.1 LTS _Noble Numbat_ - Release amd64 (20240827.1) noble Release  
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
508 packages can be upgraded. Run 'apt list --upgradable' to see them.  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
The following packages were automatically installed and are no longer required:  
  libllvm17t64 python3-netifaces  
Use 'sudo apt autoremove' to remove them.  
The following NEW packages will be installed:  
  libllvm19 libmalcontent-0-0 linux-headers-6.11.0-21-generic linux-hwe-6.11-headers-6.11.0-21  
  linux-hwe-6.11-tools-6.11.0-21 linux-image-6.11.0-21-generic linux-modules-6.11.0-21-generic  
  linux-modules-extra-6.11.0-21-generic linux-tools-6.11.0-21-generic mesa-libgallium  
The following upgrades have been deferred due to phasing:  
  ubuntu-drivers-common  
The following packages will be upgraded:  
  acl alsa-ucm-conf amd64-microcode apparmor apport apport-core-dump-handler apport-gtk base-files bind9-dnsutils  
  bind9-host bind9-lbns bluez bluez-cups bluez-obexd bpftool bsdtar bsdextrautils bsdtarutils bubblewrap cloud-init cpp-13  
  cpp-13-x86-64-linux-gnu cryptsetup cryptsetup-bin cryptsetup-initramfs cups cups-browsed cups-bsd cups-client  
  cups-common cups-core-drivers cups-daemon cups-ipp-utils cups-ppdc cups-server-common dirmngr distro-info-data  
  dmidecode dmidecode dnsmasq dns-root-data dracut-install eject evince evince-common fdisk firmware-sof-signed  
  fonts-noto-color-emoji fonts-opensymbol fwupd gcc-13-base gcc-14-base gdm3 ghostscript gir1.2-gdm-1.0  
  gir1.2-glib-2.0 gir1.2-gnomebluetooth-3.0 gir1.2-gst-plugins-base-1.0 gir1.2-gstreamer-1.0 gir1.2-gtk-3.0  
  gir1.2-gtk-4.0 gir1.2-javascriptcoregtk-4.1 gir1.2-javascriptcoregtk-6.0 gir1.2-mutter-14 gir1.2-nm-1.0
```

```
ubuntu@LikhithaUbuntu: ~  
Setting up xfsprogs (6.6.0-1ubuntu2.1) ...  
update-initramfs is disabled since running on read-only media  
Setting up ubuntu-report (1.7.3ubuntu0.24.04.2) ...  
Setting up xwayland (2:23.2.6-1ubuntu0.4) ...  
Setting up thermald (2.5.6-2ubuntu0.24.04.2) ...  
Setting up ubuntu-desktop (1.539.2) ...  
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...  
Processing triggers for rsyslog (8.2312.0-3ubuntu9.1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...  
Processing triggers for desktop-file-utils (0.27-2build1) ...  
ubuntu@LikhithaUbuntu:~$ sudo apt install apache2 mysql-server python3 python3-pip php libapache2-mod-php -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
apache2 is already the newest version (2.4.58-1ubuntu8.6).  
mysql-server is already the newest version (8.0.41-0ubuntu0.24.04.1).  
python3 is already the newest version (3.12.3-0ubuntu2).  
The following package was automatically installed and is no longer required:  
  python3-netifaces  
Use 'sudo apt autoremove' to remove it.  
The following additional packages will be installed:  
  binutils binutils-common binutils-x86-64-linux-gnu build-essential dpkg-dev fakeroot g++ g++-13  
  g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu  
  libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libapache2-mod-php8.3 libasan8 libbinutils  
  libcc1-0 libctf-nobfd0 libctf0 libdpkg-perl libexpat1-dev libfakeroot libfile-fcntllock-perl libgcc-13-dev  
  libgprofng0 libhwasan0 libitm1 libjs-sphinxdoc libjs-underscore liblsan0 libpython3-dev libpython3.12-dev  
  libquadmath0 libstdc++-13-dev libtsan2 libubsan1 lto-disabled-list make php-common php8.3 php8.3-cli  
  php8.3-common php8.3-opcache php8.3-readline python3-dev python3-setuptools python3-wheel python3.12-dev zlib1g-dev  
Suggested packages:  
  binutils-doc gprofng-gui debian-keyring g++-multilib g++-13-multilib gcc-13-doc gcc-multilib autoconf automake  
  libtool flex bison gcc-doc gcc-13-multilib gcc-13-locales gdb-x86-64-linux-gnu php-pear git bzr libstdc++-13-doc
```

```
ubuntu@LikhithaUbuntu: ~/nas-server  
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo apt install libmysqlclient-dev default-libmyclient-dev build-essential p  
kg-config python3-dev -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
E: Unable to locate package default-libmyclient-dev  
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo apt install libmysqlclient-dev build-essential pkg-config python3-dev -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
build-essential is already the newest version (12.10ubuntu1).  
build-essential set to manually installed.  
python3-dev is already the newest version (3.12.3-0ubuntu2).  
python3-dev set to manually installed.  
The following package was automatically installed and is no longer required:  
  python3-netifaces  
Use 'sudo apt autoremove' to remove it.  
The following additional packages will be installed:  
  libmysqlclient21 libpkgconf3 libssl-dev libzstd-dev pkgconf pkgconf-bin  
Suggested packages:  
  libssl-doc  
The following NEW packages will be installed:  
  libmysqlclient-dev libmysqlclient21 libpkgconf3 libssl-dev libzstd-dev pkg-config pkgconf pkgconf-bin  
0 upgraded, 8 newly installed, 0 to remove and 1 not upgraded.  
Need to get 5615 kB/5690 kB of archives.  
After this operation, 30.8 MB of additional disk space will be used.  
Get:1 cdrom://Ubuntu 24.04.1 LTS _Noble Numbat_ - Release amd64 (20240827.1) noble/main amd64 libpkgconf3 amd64 1.8.1-2b  
uild1 [30.7 kB]  
Get:2 cdrom://Ubuntu 24.04.1 LTS _Noble Numbat_ - Release amd64 (20240827.1) noble/main amd64 pkgconf-bin amd64 1.8.1-2b  
uild1 [20.7 kB]  
Get:3 cdrom://Ubuntu 24.04.1 LTS _Noble Numbat_ - Release amd64 (20240827.1) noble/main amd64 pkgconf amd64 1.8.1-2bui  
ld1 [16.8 kB]  
Get:4 cdrom://Ubuntu 24.04.1 LTS _Noble Numbat_ - Release amd64 (20240827.1) noble/main amd64 pkg-config amd64 1.8.1-2bui
```

```
Apr 11 05:32
ubuntu@LikhithaUbuntu: ~/nas-server

Setting up pkg-config:amd64 (1.8.1-2build1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ pip install flask flask-mysqldb
Collecting flask
  Using cached Flask-3.1.0-py3-none-any.whl.metadata (2.7 kB)
Collecting flask-mysqldb
  Using cached Flask_MySQLdb-2.0.0-py3-none-any.whl.metadata (3.0 kB)
Collecting Werkzeug>=3.1 (from flask)
  Using cached werkzeug-3.1.3-py3-none-any.whl.metadata (3.7 kB)
Collecting Jinja2>=3.1.2 (from flask)
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>=2.2 (from flask)
  Using cached itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click>=8.1.3 (from flask)
  Using cached click-8.1.8-py3-none-any.whl.metadata (2.3 kB)
Collecting blinker>=1.9 (from flask)
  Using cached blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting mysqlclient>=2.2.0 (from flask-mysqldb)
  Using cached mysqlclient-2.2.7.tar.gz (91 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting MarkupSafe>=2.0 (from Jinja2>=3.1.2->flask)
  Downloading MarkupSafe-3.0.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.0 kB)
  Downloading flask-3.1.0-py3-none-any.whl (102 kB)
    103.0/103.0 kB 1.4 MB/s eta 0:00:00
  Downloading Flask_MySQLdb-2.0.0-py3-none-any.whl (4.9 kB)
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
  Downloading click-8.1.8-py3-none-any.whl (98 kB)
    98.2/98.2 kB 4.2 MB/s eta 0:00:00
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
```

```
Apr 11 05:35
ubuntu@LikhithaUbuntu: ~/nas-server

  Downloading MarkupSafe-3.0.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.0 kB)
  Downloading flask-3.1.0-py3-none-any.whl (102 kB)
    103.0/103.0 kB 1.4 MB/s eta 0:00:00
  Downloading Flask_MySQLdb-2.0.0-py3-none-any.whl (4.9 kB)
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
  Downloading click-8.1.8-py3-none-any.whl (98 kB)
    98.2/98.2 kB 4.2 MB/s eta 0:00:00
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
    134.9/134.9 kB 13.0 MB/s eta 0:00:00
  Downloading werkzeug-3.1.3-py3-none-any.whl (224 kB)
    224.5/224.5 kB 12.8 MB/s eta 0:00:00
  Downloading MarkupSafe-3.0.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (23 kB)
Building wheels for collected packages: mysqlclient
  Building wheel for mysqlclient (pyproject.toml) ... done
  Created wheel for mysqlclient: filename=mysqlclient-2.2.7-cp312-cp312-linux_x86_64.whl size=129070 sha256=47c2265bc5b03dec3a1fc773b73c2b3576078b7b0a90643b50329b5a7f64cdb4
  Stored in directory: /home/ubuntu/.cache/pip/wheels/27/95/18/7f176fffd46629e710c04c810b9c4d7d4358fe7c96a7d2306d
Successfully built mysqlclient
Installing collected packages: mysqlclient, MarkupSafe, itsdangerous, click, blinker, Werkzeug, Jinja2, flask, flask-mysqldb
Successfully installed Jinja2-3.1.6 MarkupSafe-3.0.2 Werkzeug-3.1.3 blinker-1.9.0 click-8.1.8 flask-3.1.0 flask-mysqldb-2.0.0 itsdangerous-2.2.0 mysqlclient-2.2.7
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo systemctl start apache2
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo systemctl enable mysql
Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo systemctl start mysql
(venv) ubuntu@LikhithaUbuntu:~/nas-server$
```



```
Apr 11 05:37
ubuntu@LikhithaUbuntu: ~/nas-server

Installing collected packages: mysqlclient, MarkupSafe, itsdangerous, click, blinker, Werkzeug, Jinja2, flask, flask-mysqldb
Successfully installed Jinja2-3.1.6 MarkupSafe-3.0.2 Werkzeug-3.1.3 blinker-1.9.0 click-8.1.8 flask-3.1.0 flask-mysqldb-2.0.0 itsdangerous-2.2.0 mysqlclient-2.2.7
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo systemctl start apache2
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo systemctl enable mysql
Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo systemctl start mysql
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: y

There are three levels of password validation policy:

LOW      Length >= 8
MEDIUM  Length >= 8, numeric, mixed case, and special characters
STRONG Length >= 8, numeric, mixed case, special characters and dictionary      file

Please enter 0 = LOW, 1 = MEDIUM and 2 = STRONG: A0Steam@3
```

```
Apr 11 05:41
ubuntu@LikhithaUbuntu: ~/nas-server

By default, MySQL comes with a database named 'test' that
anyone can access. This is also intended only for testing,
and should be removed before moving into a production
environment.

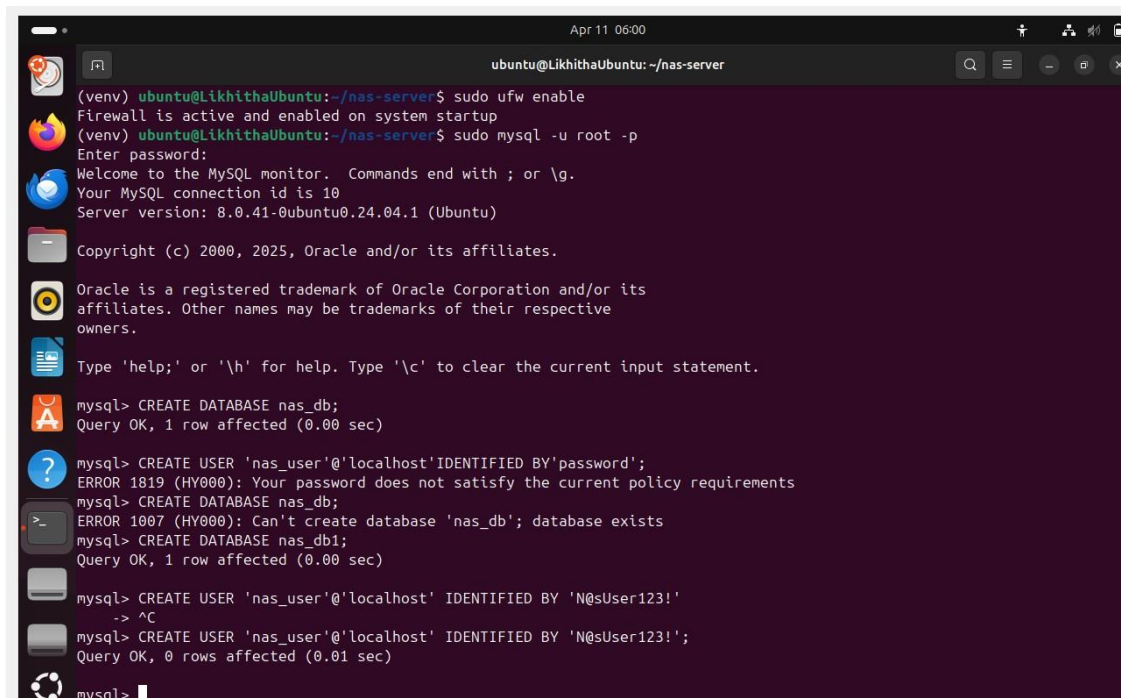
Remove test database and access to it? (Press y|Y for Yes, any other key for No) : Y
- Dropping test database...
Success.

- Removing privileges on test database...
Success.

Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : Y
Success.

All done!
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo ufw allow 22/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo ufw allow 5000/tcp
Rule added
Rule added (v6)
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo ufw enable
Firewall is active and enabled on system startup
(venv) ubuntu@LikhithaUbuntu:~/nas-server$
```

A terminal window titled 'ubuntu@LikhithaUbuntu: ~/nas-server' with a timestamp of 'Apr 11 06:00'. The terminal shows the following commands and output:

```
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo ufw enable
Firewall is active and enabled on system startup
(venv) ubuntu@LikhithaUbuntu:~/nas-server$ sudo mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.41-0ubuntu0.24.04.1 (Ubuntu)

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE nas_db;
Query OK, 1 row affected (0.00 sec)

mysql> CREATE USER 'nas_user'@'localhost' IDENTIFIED BY 'password';
ERROR 1819 (HY000): Your password does not satisfy the current policy requirements
mysql> CREATE DATABASE nas_db;
ERROR 1007 (HY000): Can't create database 'nas_db'; database exists
mysql> CREATE DATABASE nas_db1;
Query OK, 1 row affected (0.00 sec)

mysql> CREATE USER 'nas_user'@'localhost' IDENTIFIED BY 'N@sUser123!';
-> ^C
mysql> CREATE USER 'nas_user'@'localhost' IDENTIFIED BY 'N@sUser123!';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

Develop the Web Interface:

As for developing a web interface we have used the flask interface to integrate the webserver

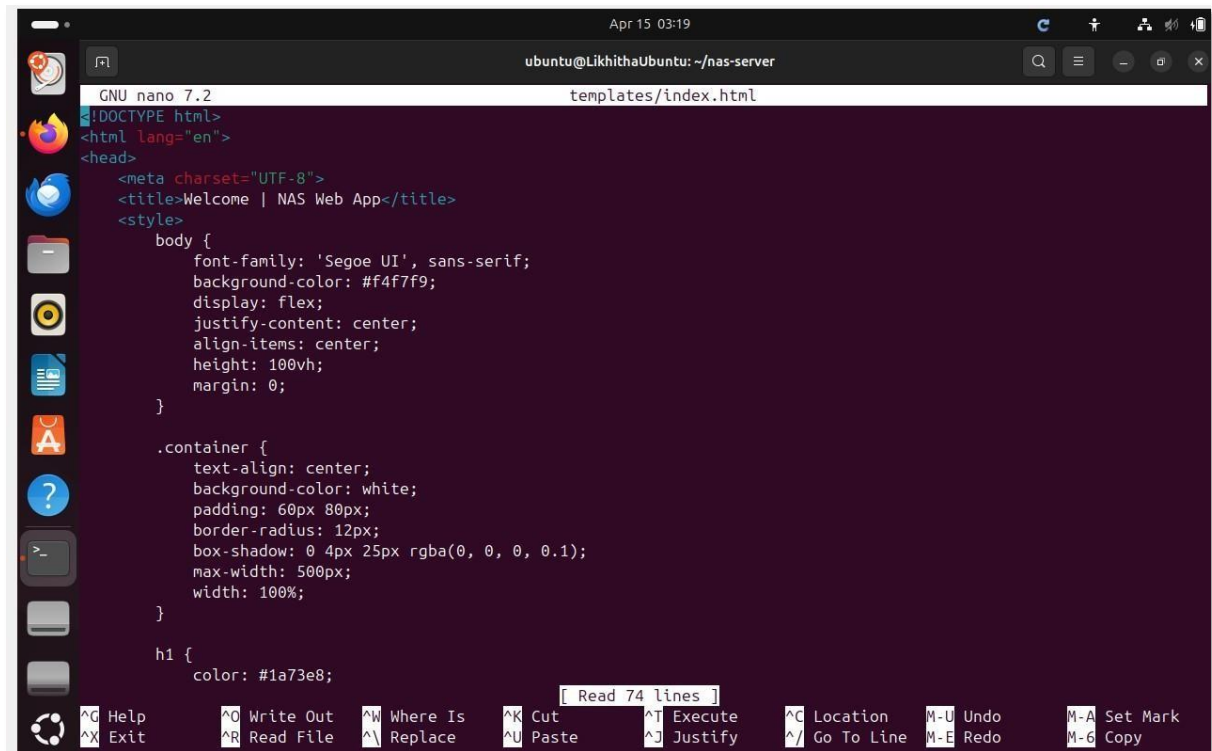
Here we have divided the overall structure into two one as templates and the app code.

In the templates we have included the html codes for the login page, dashboard, users, index

- **Index.html** contain the display config for when you open the web interface
- **Login.html** contains the setting for when you have to enter your credentials and get into the app
- **Dashboard.html** contains the config for Managing files and users once you login to the interface
- **Users.html** contains the display configs for adding or deleting users

Lastly, **app.py** is the file that contains how the app is performing and code for how the actions are done.

Index.html



Apr 15 03:19

ubuntu@LikhithaUbuntu: ~/nas-server

GNU nano 7.2 templates/index.html

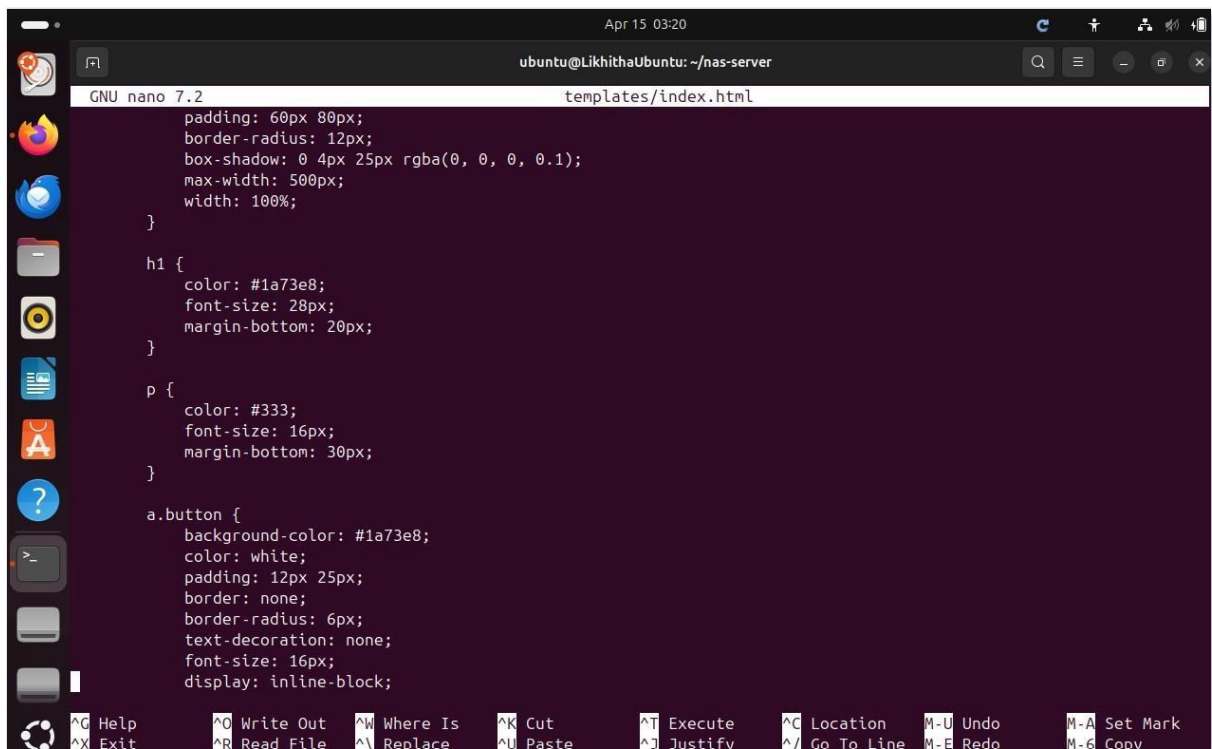
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Welcome | NAS Web App</title>
  <style>
    body {
      font-family: 'Segoe UI', sans-serif;
      background-color: #f4f7f9;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }

    .container {
      text-align: center;
      background-color: white;
      padding: 60px 80px;
      border-radius: 12px;
      box-shadow: 0 4px 25px rgba(0, 0, 0, 0.1);
      max-width: 500px;
      width: 100%;
    }

    h1 {
      color: #1a73e8;
```

[Read 74 lines]

Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark
Exit Read File Replace Paste Justify Go To Line M-E Redo M-6 Copy



Apr 15 03:20

ubuntu@LikhithaUbuntu: ~/nas-server

GNU nano 7.2 templates/index.html

```
padding: 60px 80px;
border-radius: 12px;
box-shadow: 0 4px 25px rgba(0, 0, 0, 0.1);
max-width: 500px;
width: 100%;
}

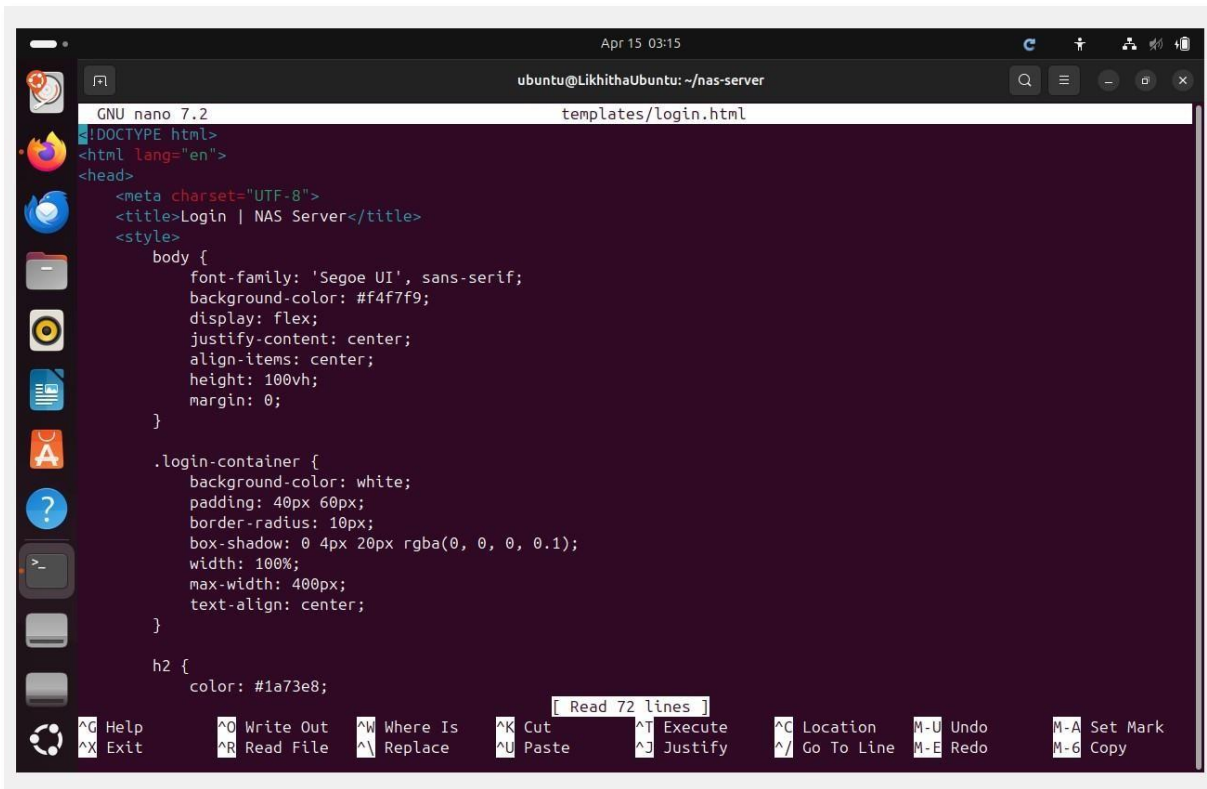
h1 {
  color: #1a73e8;
  font-size: 28px;
  margin-bottom: 20px;
}

p {
  color: #333;
  font-size: 16px;
  margin-bottom: 30px;
}

a.button {
  background-color: #1a73e8;
  color: white;
  padding: 12px 25px;
  border: none;
  border-radius: 6px;
  text-decoration: none;
  font-size: 16px;
  display: inline-block;
```

Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark
Exit Read File Replace Paste Justify Go To Line M-E Redo M-6 Copy

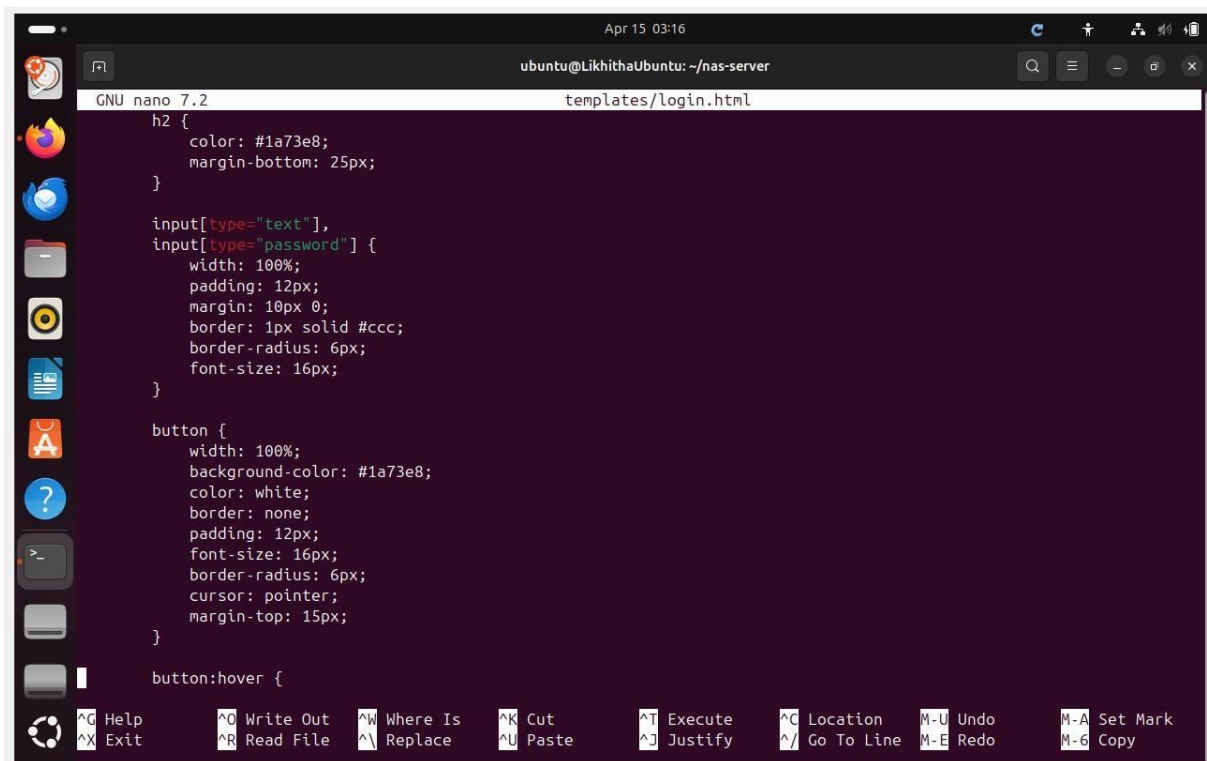
Login.html



```
GNU nano 7.2 templates/login.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Login | NAS Server</title>
  <style>
    body {
      font-family: 'Segoe UI', sans-serif;
      background-color: #f4f7f9;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }

    .login-container {
      background-color: white;
      padding: 40px 60px;
      border-radius: 10px;
      box-shadow: 0 4px 20px rgba(0, 0, 0, 0.1);
      width: 100%;
      max-width: 400px;
      text-align: center;
    }

    h2 {
      color: #1a73e8;
    }
  </style>
</head>
<body>
</body>
</html>
```



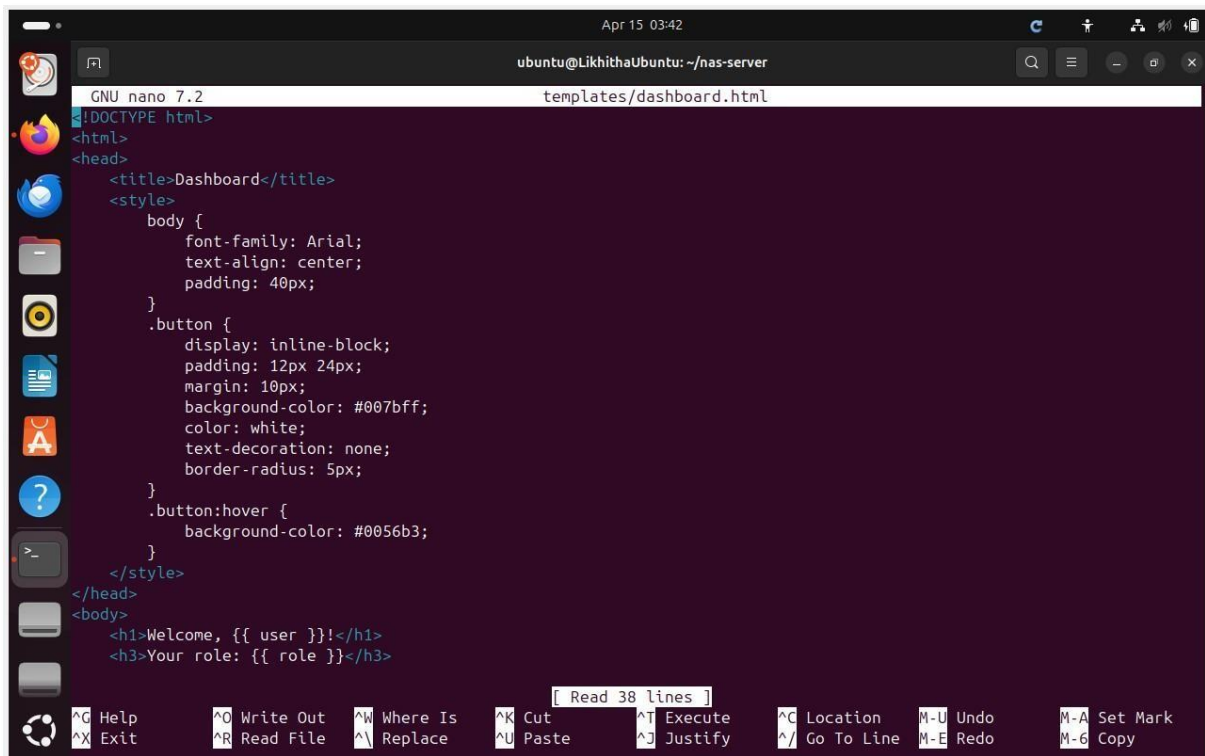
```
GNU nano 7.2 templates/login.html
h2 {
  color: #1a73e8;
  margin-bottom: 25px;
}

input[type="text"],
input[type="password"] {
  width: 100%;
  padding: 12px;
  margin: 10px 0;
  border: 1px solid #ccc;
  border-radius: 6px;
  font-size: 16px;
}

button {
  width: 100%;
  background-color: #1a73e8;
  color: white;
  border: none;
  padding: 12px;
  font-size: 16px;
  border-radius: 6px;
  cursor: pointer;
  margin-top: 15px;
}

button:hover {
  background-color: #0070c0;
}
```

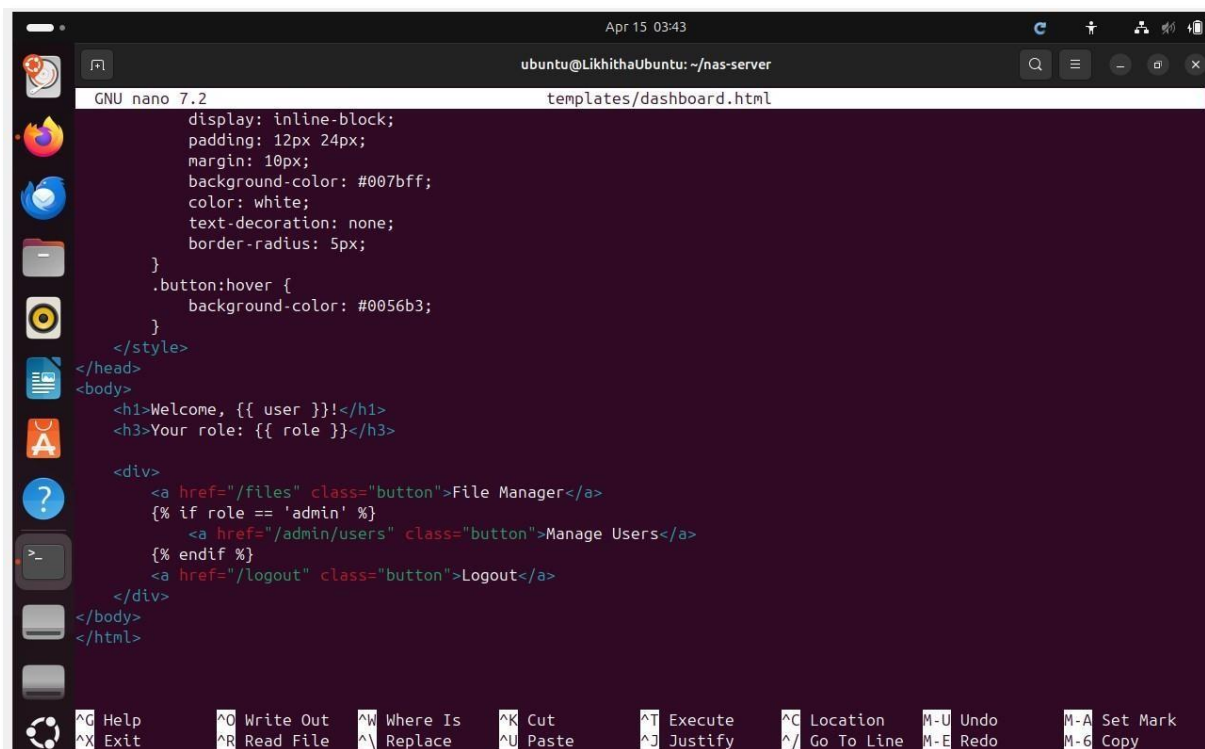
Dashboard.html



```
GNU nano 7.2 templates/dashboard.html
<!DOCTYPE html>
<html>
<head>
<title>Dashboard</title>
<style>
  body {
    font-family: Arial;
    text-align: center;
    padding: 40px;
  }
  .button {
    display: inline-block;
    padding: 12px 24px;
    margin: 10px;
    background-color: #007bff;
    color: white;
    text-decoration: none;
    border-radius: 5px;
  }
  .button:hover {
    background-color: #0056b3;
  }
</style>
</head>
<body>
<h1>Welcome, {{ user }}!</h1>
<h3>Your role: {{ role }}</h3>
```

[Read 38 lines]

Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark
Exit Read File Replace AU Paste AJ Justify G/ Go To Line M-E Redo M-6 Copy

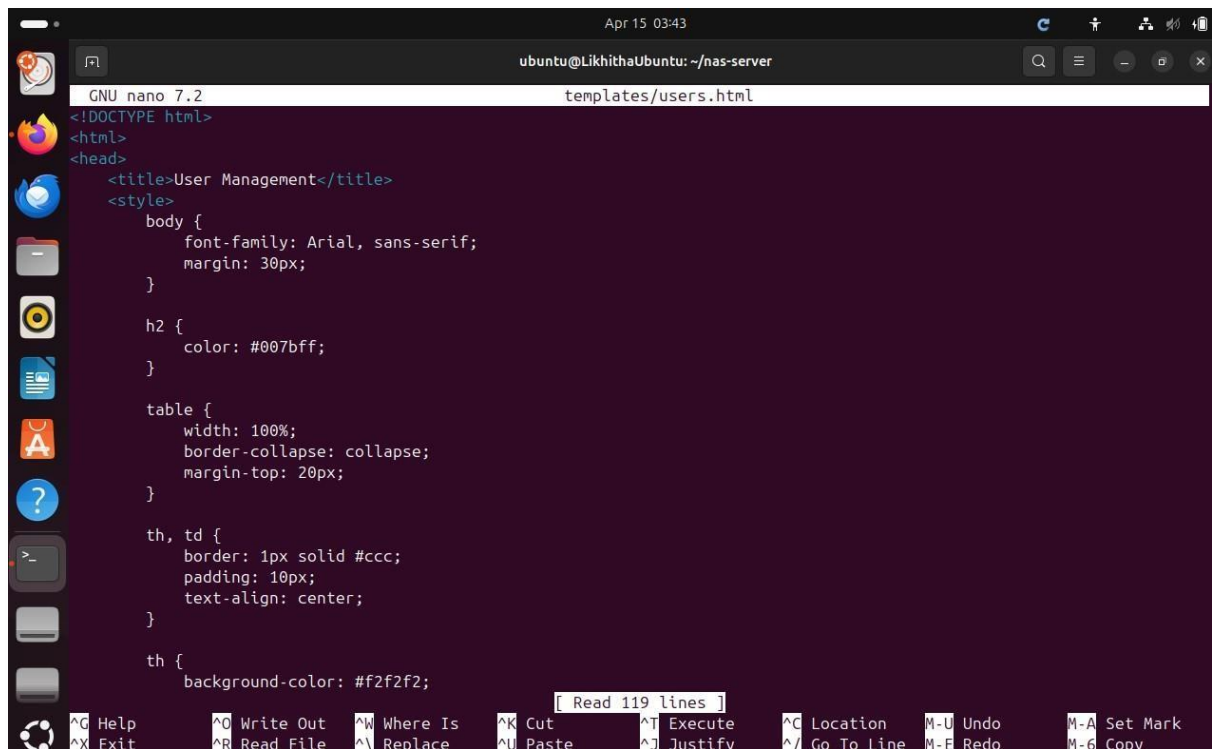


```
    display: inline-block;
    padding: 12px 24px;
    margin: 10px;
    background-color: #007bff;
    color: white;
    text-decoration: none;
    border-radius: 5px;
  }
  .button:hover {
    background-color: #0056b3;
  }
</style>
</head>
<body>
<h1>Welcome, {{ user }}!</h1>
<h3>Your role: {{ role }}</h3>

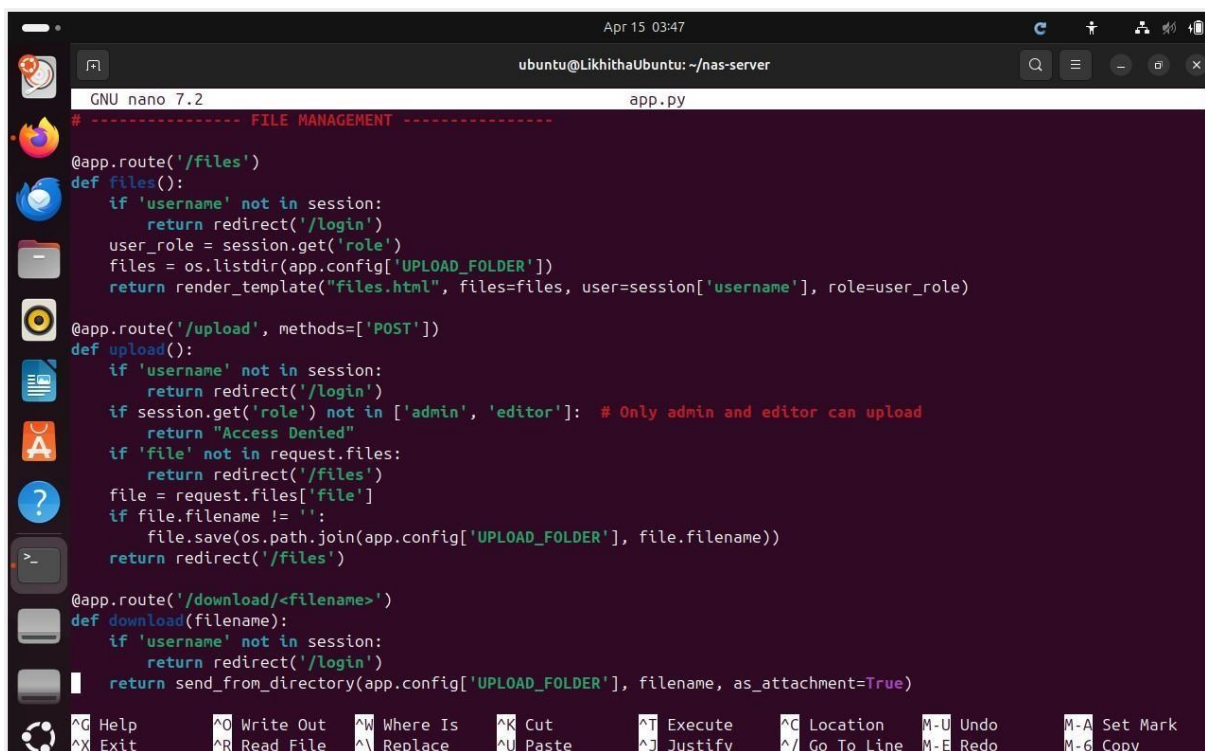
<div>
  <a href="/files" class="button">File Manager</a>
  {% if role == 'admin' %}
    <a href="/admin/users" class="button">Manage Users</a>
  {% endif %}
  <a href="/logout" class="button">Logout</a>
</div>
</body>
</html>
```

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Users.html



```
GNU nano 7.2 templates/users.html
<!DOCTYPE html>
<html>
<head>
  <title>User Management</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 30px;
    }
    h2 {
      color: #007bff;
    }
    table {
      width: 100%;
      border-collapse: collapse;
      margin-top: 20px;
    }
    th, td {
      border: 1px solid #ccc;
      padding: 10px;
      text-align: center;
    }
    th {
      background-color: #f2f2f2;
    }
  
```



```
GNU nano 7.2 app.py
# ----- FILE MANAGEMENT -----
@app.route('/files')
def files():
    if 'username' not in session:
        return redirect('/login')
    user_role = session.get('role')
    files = os.listdir(app.config['UPLOAD_FOLDER'])
    return render_template("files.html", files=files, user=session['username'], role=user_role)

@app.route('/upload', methods=['POST'])
def upload():
    if 'username' not in session:
        return redirect('/login')
    if session.get('role') not in ['admin', 'editor']: # Only admin and editor can upload
        return "Access Denied"
    if 'file' not in request.files:
        return redirect('/files')
    file = request.files['file']
    if file.filename != '':
        file.save(os.path.join(app.config['UPLOAD_FOLDER'], file.filename))
    return redirect('/files')

@app.route('/download/<filename>')
def download(filename):
    if 'username' not in session:
        return redirect('/login')
    return send_from_directory(app.config['UPLOAD_FOLDER'], filename, as_attachment=True)
```

app.py

```
GNU nano 7.2 app.py
from flask import Flask, render_template, request, redirect, session, send_from_directory
from flask_mysql import MySQL
import os
from datetime import datetime

app = Flask(__name__)
app.secret_key = 'supersecurekey123' # Required for session support

# --- MySQL Configuration ---
app.config['MYSQL_HOST'] = 'localhost'
app.config['MYSQL_USER'] = 'nas_team3'
app.config['MYSQL_PASSWORD'] = 'Aosteam@3'
app.config['MYSQL_DB'] = 'nas_db'

mysql = MySQL(app)

# --- File Upload Directory ---
UPLOAD_FOLDER = 'uploads'
os.makedirs(UPLOAD_FOLDER, exist_ok=True)
app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER

# ----- ROUTES -----

@app.route('/')
def home():
    return render_template("index.html", current_year=datetime.now().year)

[ Read 139 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo      M-6 Copy
```

```
GNU nano 7.2 app.py
@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        username = request.form['username']
        password = request.form['password']

        print("Trying login with:", username, password) # Debugging info

        cur = mysql.connection.cursor()
        cur.execute("SELECT role FROM users WHERE username=%s AND password=%s", (username, password))
        result = cur.fetchone()
        cur.close()

        if result:
            session['username'] = username
            session['role'] = result[0]
            print("Login success:", username, result[0]) # Debugging info
            return redirect('/dashboard')
        else:
            return render_template("login.html", error="Invalid credentials")

    return render_template("login.html")

@app.route('/logout')
def logout():
    session.clear()
    return redirect('/login')
```

```
ubuntu@LikhithaUbuntu: ~/nas-server
GNU nano 7.2 app.py

#----- ADMIN USER MANAGEMENT -----

@app.route('/admin/users')
def admin_users():
    if 'username' not in session or session['role'] != 'admin':
        return redirect('/login')
    cur = mysql.connection.cursor()
    cur.execute("SELECT id, username, role FROM users")
    users = cur.fetchall()
    cur.close()
    return render_template('users.html', users=users)

@app.route('/admin/users/add', methods=['POST'])
def add_user():
    if 'username' not in session or session['role'] != 'admin':
        return redirect('/login')

    username = request.form['username']
    password = request.form['password']
    role = request.form['role']

    cur = mysql.connection.cursor()
    cur.execute("INSERT INTO users (username, password, role) VALUES (%s, %s, %s)", (username, password, role))
    mysql.connection.commit()
    cur.close()

    return redirect('/admin/users')

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```

```
ubuntu@LikhithaUbuntu: ~/nas-server
GNU nano 7.2 app.py

@app.route('/admin/users/add', methods=['POST'])
def add_user():
    if 'username' not in session or session['role'] != 'admin':
        return redirect('/login')

    username = request.form['username']
    password = request.form['password']
    role = request.form['role']

    cur = mysql.connection.cursor()
    cur.execute("INSERT INTO users (username, password, role) VALUES (%s, %s, %s)", (username, password, role))
    mysql.connection.commit()
    cur.close()

    return redirect('/admin/users')

@app.route('/admin/users/delete/<int:user_id>', methods=['POST'])
def delete_user(user_id):
    cur = mysql.connection.cursor()
    cur.execute("DELETE FROM users WHERE id = %s", (user_id,))
    mysql.connection.commit()
    cur.close()
    return redirect('/admin/users')

# ----- Run Server -----
if __name__ == '__main__':
    app.run(debug=True, host='0.0.0.0', port=5000)

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```

Configure Port Forwarding

Below are the screenshots where we are allowing the 22/tcp and 5000 and other routing configurations and where we show what are all the routes allowed

```
ubuntu@LikhithaUbuntu:~$ sudo apt enable mysql
E: Invalid operation enable
ubuntu@LikhithaUbuntu:~$ sudo systemctl enable mysql
Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
ubuntu@LikhithaUbuntu:~$ sudo systemctl start mysql
ubuntu@LikhithaUbuntu:~$ sudo apt install python3 python3-flask
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2).
python3-flask is already the newest version (3.0.2-1ubuntu1).
The following package was automatically installed and is no longer required:
  python3-netifaces
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
ubuntu@LikhithaUbuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@LikhithaUbuntu:~$ sudo ufw allow 22/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
ubuntu@LikhithaUbuntu:~$ sudo ufw status
Status: active
```

```
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
ubuntu@LikhithaUbuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@LikhithaUbuntu:~$ sudo ufw allow 22/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
ubuntu@LikhithaUbuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
Apache ALLOW Anywhere
OpenSSH ALLOW Anywhere
80/tcp ALLOW Anywhere
5000/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
Apache (v6) ALLOW Anywhere (v6)
OpenSSH (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)
5000/tcp (v6) ALLOW Anywhere (v6)

ubuntu@LikhithaUbuntu:~$
```


Implement User Access Control:

Below are some of the screenshots where we implement different admin configurations like creating a user adding a user and giving access restrictions

Also, for file management like uploading and deleting files

Here we have should an example where we created a user with only reader access and how the user didn't get any option to upload files

